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# BENDING WITHIN THE AXIAL TECTONIC BELT OF NEW ZEALAND IN THE LAST 9 MYR FROM PALEOMAGNETIC DATA

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Tongaporutuan Stage (Late Miocene, 11–6 Myr) sedimentary rocks at two localities within the axial tectonic belt of New Zealand have a direction of magnetisation in which the declination is 020 ± 11° and 030 ± 8° ($D \pm \Delta_{95}$). The directions of magnetisation in rocks of similar age outside the belt have a declination like that of the earth's present axial dipole, 360°. The clockwise deviation of the declination is inferred to indicate bending of the rocks within the zone of deformation between the Pacific and Indian plates due to the oblique convergence of these plates.

## 1. Introduction

In central parts of the South Island of New Zealand the relative motion between Pacific and Indian plates is being taken up in a 100-km-wide belt of deformation, in the axial tectonic belt, which is bounded on the northwest by the Alpine Fault and which grades southeastward into the undeformed Pacific plate [1]. The belt continues northeastward across Cook Strait into the eastern North Island where the Pacific plate is being subducted at the Hikurangi Trench (Fig. 1). In the South Island and southern part of the North Island the direction of the principal axis of compression from both geodetic and earthquake mechanism data is about 110° [1,2]. The trend of the belt is 050° so that the mode of deformation is a predominant clockwise simple shear parallel to the belt with a lesser normal compressional component. The strain rate from repeated triangulation surveys in the last 100 years averaged across the 100 km width of the belt is $5 \times 10^{-7}$/yr, about the same as inferred from the plate tectonic rate given by the Euler poles of Minster and Jordan [3] and Chase [4].

Within the axial tectonic belt south of Cook Strait are four major transcurrent faults striking at about 60–70° that offset Late Quaternary river terraces. The total amount of offset on these faults, however, is only about one third the calculated relative plate motion over the same period. The average rate of displacement across the full width of the belt in the last 18,000 years from offset terraces is 16 ± 2 mm/yr [5], whereas the plate tectonic rate is 50 mm/yr, similar to the inferred rates from geodetic data for the last 100 years.

This discrepancy suggests that only part of the plate motion is taken up by movement on these faults and much of the movement may be occurring as a deformation in the rocks between the faults. If this takes place as a simple shear then reference lines inclined to the direction of shear will be bent. With the rates estimated from geodetic and plate tectonic data the amount of bending could be very substantial; rates of $3 \times 10^{-7}$/yr simple shear continued for 8 Myr will result in a 67° rotation of a line initially perpendicular to the shear.

As the axial tectonic belt is approached from the east (the Pacific plate) the strike of lithologic bound-

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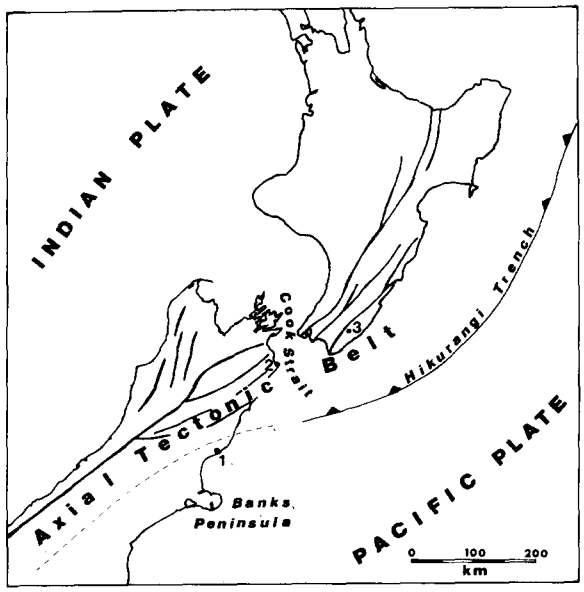



Fig. 1. Location map. Paleomagnetic data are available from Banks Peninsula [10], and localities *1* (Motunau River), *2* (Cape Campbell) and *3* (Hinakura).

aries and bedding swings through about 90°; in the undeformed Pacific plate schist and greywacke strike northwest while in the axial tectonic belt they strike northeast. It has been suggested that this bending is caused by relative plate motion in Late Cenozoic times and is continuing today [1].

With the view of testing this hypothesis, oriented cores of sedimentary rock of Late Tertiary age were collected for a paleomagnetic study. If bending has occurred since the Late Tertiary then the direction of magnetisation in rocks from the axial tectonic belt should show a clockwise rotation compared to the direction in rocks of similar age in the undeformed plate.

## 2. Sample collection

Oriented cores of Late Miocene sedimentary rocks were collected at the three locations shown in Fig. 1. The rocks range from Tongaporutuan to Opoitian stages in the New Zealand nomenclature. The age equivalents of the New Zealand stages are: Tongaporutuan (11–6 Myr), Kapitean (6–5 Myr) and Opoitian (5–3 Myr) (7). Each section is described below where the numbers in parentheses refer to the location diagram of Fig. 1.

### *2.1. Motunau River section (locality 1)*

The section collected is the stratigraphically highest part of the section described by Wilson [6] in the lower Motunau River. Two sequences were collected. The first from a calcareous sandstone below the base of the Tongaporutuan at approximately 3-m intervals upward through the section to an Opoitian limestone and conglomerate (numbers 54–85, Fig. 2). The second, at about 1-m intervals, over the lower half of the previous section (numbers 226–269, Fig. 2). In the lower part of the section the rocks are massive grey, and blue-grey siltstones; bedding is uncommon. Between samples 70 and 82 the rocks are banded siltstones and mudstones.

Six samples were collected for fossil identification of age and were examined by Dr. George Scott of the New Zealand Geological Survey:

Locality 3: Pre-Tongaporutuan – the unit is equivalent to the Altonian Stage in Wilson [6].

4: Tongaporutuan, probably lower, with a good fauna from deep water facies.

5, 6: Like the succeeding samples these have a shallow-water marine assemblage. Stratigraphic control is difficult because of the poor representation of planktonic species. Upper Tongaporutuan to Kapitean.

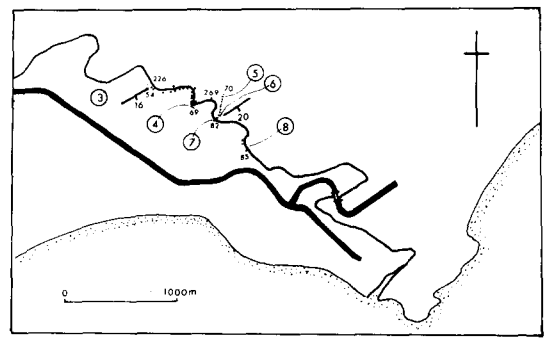



Fig. 2. Motunau River section. The dots and small numbers are the location and numbers of samples collected for paleomagnetic study. The numbers in circles are the samples collected for fossil age determination. Dip and strike of the bedding are also shown.

7: A specimen of *Globorotalia conomiozea* suggests Kapitean.

8: *Globorotalia puncticulata* is present. Opoitian.

### 2.2. *Cape Campbell sections (locality 2)*

Between Cape Campbell and Mussel Point is exposed a 1350-m-thick sequence of largely Tongaporutuan sedimentary rocks folded into an asymmetric syncline (Fig. 3). The rocks consist of bedded siltstones and fine-grained sandstones, massive grey siltstones and, in one place, thick turbidites. The turbidite beds, each about 500 mm thick, have grit or coarse sandstone base with a mudstone upper part about 50–100 mm thick and, overall, are about 20 m thick. Infraformational breccias with stretched out mudstone clasts, particularly well developed near the lighthouse, contorted bedding in some of the apparently massive siltstones and internal folds within the sequence indicate considerable deformation and slumping during deposition. The large thickness of rocks of a restricted age and the occurrence of turbidites suggests rapid deposition in relatively deep water. The fine mudstones in the thick turbidite sequence are the only parts of the sections that indicate short periods of slow sedimentation during the overall vigorous redeposition of the section.

Six samples were collected for fossil age determination. The locations (numbers 10–15) are shown in Fig. 3 and the identifications by Dr. George Scott are:

Locality 10: Tongaporutuan, probably low.

11: lacks *Bolivinita*, main criterion of Tongaporutuan but fauna consistent with this stage.

12: *Bolivinita pohana* (advanced) present. Tongaporutuan.

13, 14, 15: *Globoquadrina dehiscens.* Tongaporutuan

All fauna are fairly sparse and probably represent deep-water facies.

We recognise four separate sequences in this locality. The thick turbidites sequence (numbers 196–225), coastal sequence (86–122), Fisherman's Creek sequence (159–195), and the Lighthouse sequence (601–618).

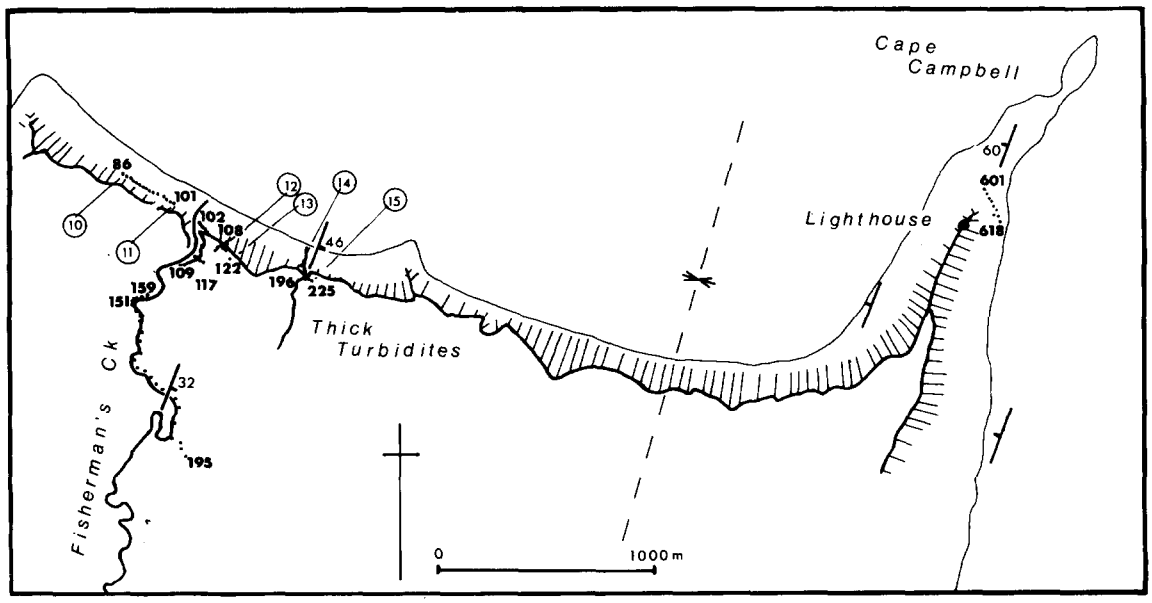



Fig. 3. Cape Campbell sections. The position of the syncline axis is approximate. Fossil age sample numbers in circles, paleomagnetic sample numbers and location shown by small numbers and dots.

### *2.3. Hinakura section (locality 3)*

Lienert et al. [8] obtained geomagnetically determined dates on a section of Uppper Miocene and Pliocene rocks exposed on the Hinakura Road in southeastern North Island. The base of the sequence was inferred to be Upper Tongaporutuan in age. We collected the lowest part of the sequence along the road and also a sequence exposed in the Pahaoa River bend adjacent to Hinakura School (Fig. 4). The collected sequence is apparently stratigraphically lower than that described by Lienert et al. but a fault, shown on the 1 : 250,000 geological map of Wellington [9], probably cuts across the road between the river and the road sequences.

The river part of the section is very well exposed and consists of well bedded grit and sandstones with minor, thin, interbedded mudstones. The upper part of the sequence is capped with a rhyolitic tuff. Samples of the mudstone were collected at approximately 3-m intervals over a total stratigraphic thickness 50 m.

The road section is very poorly exposed and bedding was not observed. Slumping on the steep sides of the valley up which the road runs is observed and there is considerable uncertainty about the tectonic correction for these samples. None gave a stable magnetisation and only the river samples are used here.

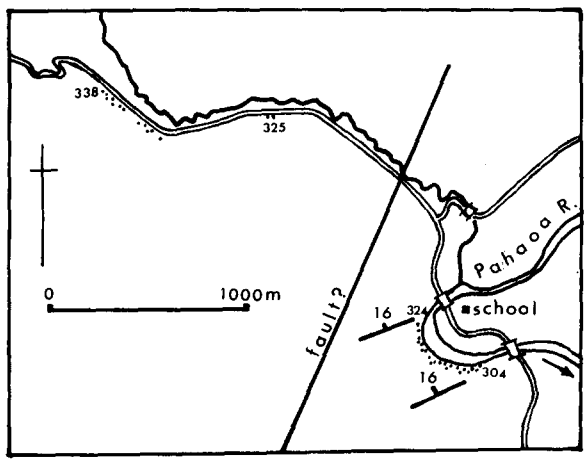



Fig. 4. Hinakura section. The position of the possible fault from Kingma [9]. No dip and strike available for the section along the road.

## 3. Paleomagnetic measurements

The 242 oriented cores collected were cut into 25-mm-long cylinders and yielded 333 samples. The direction and intensity of magnetisation were measured for NRM and again after thermal cleaning at 180°C, 230°C and 280°C on the Australian National University cryogenic magnetometer. Eight pairs of measurements were made on each occasion and were combined to give two independent estimates of the direction of magnetisation in the sample. Where these differed by more than 20°, invariably where the magnetisation is very weak, the value was rejected as meaningless.

Zijderveld plots were used to study the stability of magnetisation of each sample. All samples showed a relatively strong viscous component that was largely removed after the 180°C or 230°C cleanings. Where the direction changed between the 230°C and 280°C steps the sample was rejected as showing no significantly single stable magnetisation. It was not possible to remeasure at higher temperatures.

Only 147 samples exhibited an apparently persistent stable magnetisation after the 230°C and 280°C cleanings. The intensity of magnetisation of the rejected samples at the higher temperature cleaning was generally very low, less than $3 \times 10^{-4}$ A/m.

The intensities of magnetisation of the apparently stable samples in all sections range from $4 \times 10^{-4}$ A/m to more than $2 \times 10^{-3}$ A/m.

Although samples were collected as nearly as possible at regular horizontal intervals through each section, those samples with stable magnetisation tend to be clustered together, and there are large parts of each section showing no stable magnetisation. This prohibits clear identification of the magnetic stratigraphy and in no section is it possible to identify unambiguously a sequence of reversed and normally magnetised beds that may be compared with the established magnetic stratigraphy. Although the rocks on fossil and mapping evidence are mostly Tongaporutuan stage, and hence within the range 6–11 Myr, their precise location within this interval has not been determined.

Only 14% of the samples have reversed magnetisation. This, however, is to be expected as more than 80% of the period between 6 and 11 Myr was of normal magnetisation.

### 3.1. Results

The directions of the stable magnetisation at the 280°C cleaning are plotted on equal area nets (Fig. 5) and the results of the statistical analyses are given in Table 1. This table also includes the results of a paleomagnetic study by Evans [10] of the Akaroa sequence of volcanic rocks on Banks Peninsula (Fig. 1) that are 8–9 Myr old by potassium-argon dating.

Many of the samples are only weakly magnetised and noise, contributed by the demagnetising and measuring systems, is likely to introduce a significant scatter to their directions. Accordingly the data in each group are divided, and plotted separately, in two sub-groups of those greater and those less than $10^{-3}$ A/m.

At Motunau (Fig. 5A) 29 samples are accepted on the stability criteria of which 11 are in the more intensely magnetised sub-group. The mean direction (Table 1) is closely similar to that of the present axial dipole field.

The thick turbidite section gives the least scatter of the data collected at Cape Campbell and all but 7 of the 40 samples collected are stably magnetised. The well determined mean direction shows a 20° clockwise deviation of the declination.

Both the coastal and Fisherman's Creek section (Fig. 5C and D) show a large scatter in directions of magnetisation but again the mean declination deviates clockwise with respect to the present axial dipole field. There was poor control of the bedding in both of these sections as samples were collected over quite long distances in apparently massive siltstone. Intraformational contortions are observed in parts of the section and slumping into the creek is a possibility. Thus it is likely that the scatter is due to incorrectly estimated strike and dip. We do not place much weight on the results from these sections.

The results of the Lighthouse section at Cape Campbell are very different from the others. Only 7 samples of this section are stably magnetised through to 280°C. The directions shown crossed in Fig. 5E are from the 230° cleaning when the Zijderveld plots suggest stability. The mean direction (Table 1) is horizontal and perpendicular to the strike of the beds but

TABLE 1

Summary of the results of the statistical analyses

| Section | Group | $N$ | $\bar{D}$ (°) | $\bar{I}$ (°) | $k$ | $\alpha_{95}$ (°) | $\Delta_{95}$ (°) |
|---|---|---|---|---|---|---|---|
| Banks Peninsula * | normal | 23 | 004.2 | −60.7 | | 9.3 | 19.2 |
| | reversed | 47 | 174.7 | 52.1 | | 5.3 | 8.6 |
| | all | 70 | 357.3 | −54.9 | | 4.7 | 8.2 |
| Motunau | $J > 10^{-3}$ A/m | 11 | 354.3 | −61.7 | 17.3 | 10.1 | 21.7 |
| | all | 36 | 353.7 | −57.8 | 6.7 | 9.0 | 17.1 |
| Cape Campbell | | | | | | | |
| (a) Turbidites | $J > 10^{-3}$ A/m | 12 | 024.4 | −60.5 | 56.2 | 5.4 | 11.0 |
| | all | 33 | 020.0 | −65.6 | 27.0 | 4.7 | 11.4 |
| (b) Coastal | $J > 10^{-3}$ A/m | 12 | 032.2 | −67.7 | 24.1 | 8.2 | 22.1 |
| | all | 18 | 028.7 | −67.2 | 19.4 | 7.5 | 19.7 |
| (c) Fisherman's Creek | $J > 10^{-3}$ A/m | 12 | 031.3 | −78.5 | 15.4 | 10.3 | 63.7 |
| (d) Lighthouse | all | 24 | 124.3 | 02.3 | 2.9 | 16.8 | – |
| Hinakura | $J > 10^{-3}$ A/m | 9 | 029.4 | −63.1 | 97.3 | 4.7 | 10.4 |
| | all | 21 | 030.1 | −63.1 | 82.3 | 3.4 | 7.5 |

$N$ = number of samples, $\bar{D}$ = mean declination, $\bar{I}$ = mean inclination, $k$ = Fisher statistic, $\alpha_{95}$ = radius of the 95% confidence circle about the mean, $\Delta_{95}$ = standard error in declination = $\sin^{-1}$ ($\sin \alpha_{95}/\cos I$).

* From Evans [10, table 3].

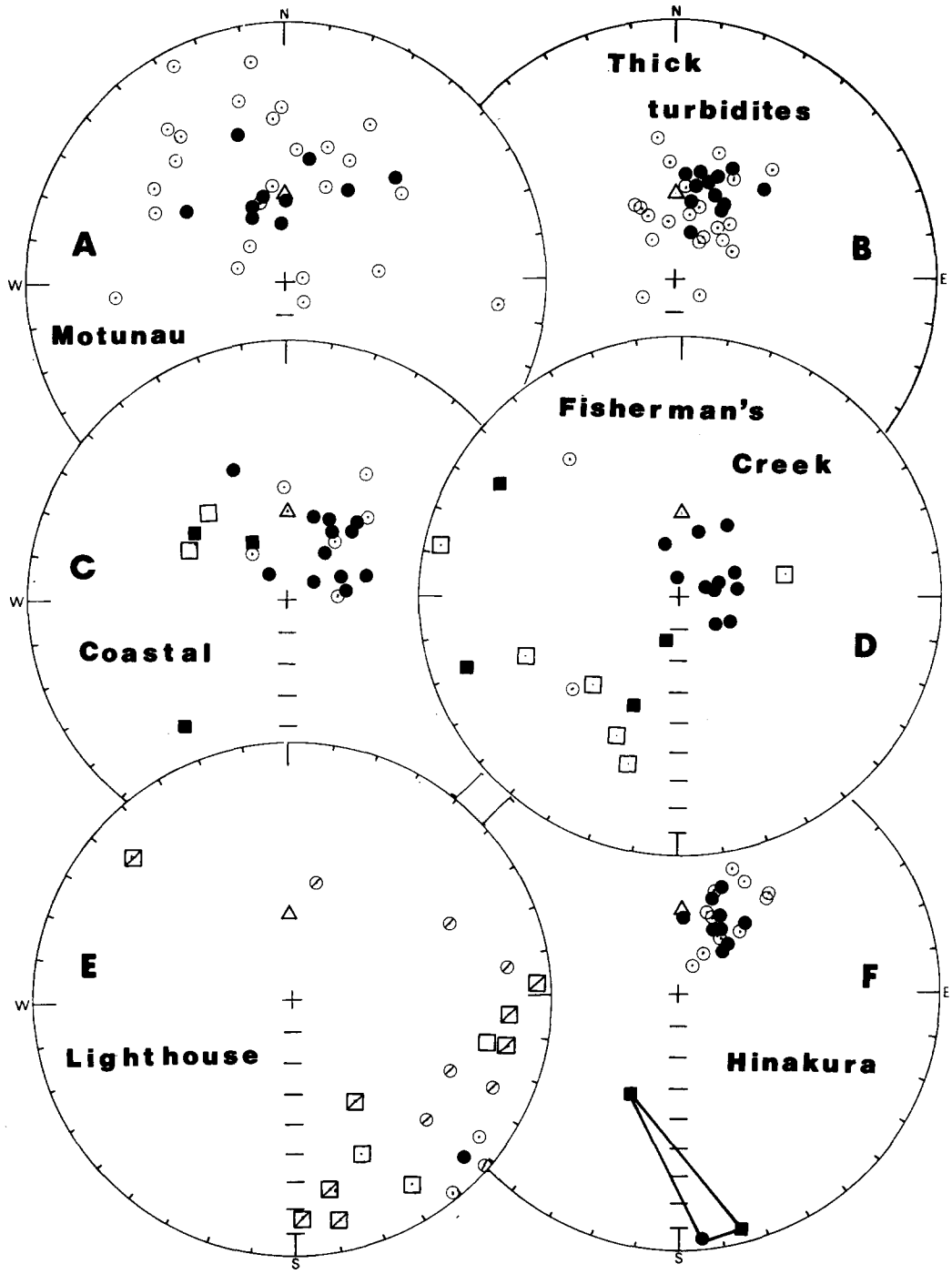



Fig. 5. Directions of magnetisation. Corrections for bedding dip applied by rotation about the strike. Solid and open symbols represent, respectively, samples with intensities greater and less than $10^{-3}$ A/m after the 280°C cleaning. Circles are normal, and squares reverse, magnetisation. The triangle is the direction of a present-day dipole aligned along the rotation axis. The crossed symbols in E are directions after the 230° cleaning – these samples being unsuitable after the 280°C cleaning step. The symbols connected by lines in F are of three specimens from the same core.

lies in the plane of bedding. The abundance of stretched out mudstone clasts and a strong foliation in the plane of bedding, indicate that these rocks have undergone intense intraformational shearing during deposition. The samples were collected from the finer-grained fractions, avoiding as much as possible locations showing macroscopic shearing. However, it is believed that the anomalous directions of magnetisation in this section are due to the deformation during deposition and is not related to post-tectonic rotation. They are considered unsuitable for this study.

The most consistent results were those from Hinakura exposed in the bend of the Pahaoa River. All samples collected were stably magnetised and give a well-defined mean direction in which the declination deviates 30° clockwise from the present axial dipole field. The three samples in the southwest quadrant, connected by lines, are all from the same core drilled in the tuff overlying the section. They indicate a possible paleomagnetic reversal. Samples collected from the road section at Hinakura were all unstably magnetised.

## 4. Conclusions

The direction of magnetisation produced by an axial dipole field at the latitude of Banks Peninsula, at the southernmost locality of Table 1, and at Hinakura, the northernmost, is 360° declination and −62.5° and −60.4° inclination respectively. For all localities it is given to sufficient accuracy by $D$ = 360° and $I$ = −61° and this is the direction shown by the triangles in Fig. 5. The direction appropriate for the Tongaporutuan pole however will differ because of plate movement and because of polar wander. The effect of Pacific plate movement can be estimated by allowing for its motion with respect to the Antarctic plate over the last 9 Myr. To adequate accuracy this motion is given by the Minster and Jordan [3] Pacific/Antarctic rotation of 8.8° (= 0.976 deg/Myr × 9 Myr) about a pole of 65.39°S and 100°E. Plate motion therefore has shifted the apparent dipole to latitude 86.3°S and longitude 004°E giving a direction of magnetisation at Banks Peninsula of $I$ = −65.3° and $D$ = 000.7°; the effect of Pacific plate motion is to steepen the apparent inclination without significantly affecting the declination.

However, Evans [10] obtained an inclination of −54.9 and declination of 357.3° for 70 samples from the Banks Peninsula lavas (Table 1) which gives an apparent pole of 81.3°S and 338.5°E referred to the Pacific plate. Corrected for plate motion this gives a pole referred to the Antarctic plate at 78°S and 355°E a very much larger deviation from the present-day axial dipole than that inferred from similar data in Australia. McElhinny and Wellman [11] suggest on the basis of studies of 17-Myr-old volcanics that the pole of that time lay near 81°S, 030°E referred to the Antarctic. Both studies, however, indicate an apparent movement of the pole approximately along the Greenwich Meridian. This will affect the inclination, but not significantly the declination, of the direction of magnetisation in New Zealand. Hence, the expected declination of the Tongaporutuan axial dipole field is within a few degrees of the present.

The direction of magnetisation of those sections within the axial tectonic belt, i.e. Hinakura and Cape Campbell compared with those outside, i.e. Motunau River and Banks Peninsula show a clockwise rotation significant at the 95% confidence level with means of 20 and 30°. Although there is a large scatter in the Motunau River section, particularly for the more weakly magnetised samples the average direction of declination is similar to that at Banks Peninsula and does not differ significantly from the present declination (Table 1). At Cape Campbell the most tightly determined direction is given by the more intensely magnetised sub-group of the thick turbidites with 24° declination ($\Delta_{95}$ = 11.4°) and −60.5° inclination. The data for the Coastal and Fisherman's Creek sections are similar although the scatter is considerably larger, as expected because of the comparative lack of control on the strike and dip of the beds. Hinakura section gives the tightest magnetisation direction of all with a 30° declination ($\Delta_{95}$ = 7.5°) and −63° inclination.

In order to average out secular variation a collection of samples representing a time interval greater than $10^4$ years is required [12]. Both the thick turbidite and Hinakura sequences are only a few tens of metres thick and turbidites can be deposited very rapidly. In the absence of any paleomagnetic reversal within these sequences we cannot be sure that they do represent sufficient time to avoid a bias to the

direction of the declination because of the secular variation even though they contain more than 60 and 200 individual turbidite beds, respectively. In view of the similar apparent rotation of the declination in all sequences (excepting that of the Lighthouse) it is highly unlikely that secular variation can be regarded as the cause of the apparent rotation, but it is possible that the secular variation introduces an uncertainty into the estimation of the amount of tectonic rotation for the thick turbidite and Hinakura sequences. The amount of tectonic rotation is therefore not well determined; it is probably greater than 10° and less than 50° than in the last 9 Myr.

Lithological boundaries, fold axes, the strike of bedding of greywackes and the schistosity of schists have an overall trend within the axial tectonic belt of northeast-southwest. Outside the belt in the South Island they tend northwest-southeast and further east on the Campbell Plateau and Chatham Rise the trend of magnetic lineations is more east-west. A rotation of around 30° in about 9 Myr suggests that much, if not all, of this apparent bending about vertical axes has occurred since mid-Tertiary times.

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Dr. M.W. McElhinny kindly made available the Australian National University cryogenic magnetometer and laboratory facilities. Dr. George Scott examined samples collected in the field and provided fossil age information. Dr. N. de B. Hornibrook permitted us to cite unpublished work relating New Zealand Fossil Stages to radiometric ages. Dr. C.G.A. Harrison and Dr. R. Van der Voo pointed out some weaknesses in an earlier draft. We are very grateful for all this assistance.

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